Received: 19 December 2023 | Revised: 5 March 2024 | Accepted: 3 April 2024

High Voltage

DOI: 10.1049/hve2.12470

ORIGINAL RESEARCH

10 kV nanosecond pulse generator with high voltage gain and reduced switches

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Associate Editor: Zhiyuan Liu

Funding information
National Natural Science Foundation of China, Grant/Award Numbers: 52277135, 52237010; Chongqing Graduate Student Research Innovation Project, Grant/Award Number: CYB23027

Abstract
In the article, a new type boost high-voltage nanosecond pulse generator is proposed. The distributed inductance of the transmission line is utilised as the energy storage unit and cooperated with the variable impedance transmission line transformer to generate nanosecond pulses with extremely high-voltage gain. What's more, the isolation effect caused by the transmission line time delay is applied to achieve modular stacking. The demand for charging power supply can be greatly reduced, and few switches are used. Finally, the topological principle is verified by experiments, and a prototype of the five-stage stacking prototype is built. With the charging voltage of 28 V, the generator can output pulse with a voltage amplitude of 10 kV and pulse width of 12 ns whose voltage gain is up to 357 times.

1 | INTRODUCTION

Recent years, high-voltage nanosecond pulse generators (HVNPG) have been widely used in industrial or medical fields [1, 2], such as atmospheric pressure low-temperature plasma [3], electrical tumour ablation [4, 5] and other high-voltage pulse applications.

There are several traditional types of generators that can produce HVNPG such as Marx [6], Blumlein line [7], LTD [8]. The feature of most of the topologies is to achieve high-voltage output through modular superposition [9, 10]. Since capacitors are always used as energy storage components, the output voltage of the single module is consistent with the voltage input of DC charging power supply. The output voltage gain is equal to the number of modules.

Generally, increasing the charging voltage can improve the pulse voltage [8]. However, due to the voltage resistance of commercial semiconductor switches, achieving high-voltage output requires more module stacking [11]. This means that more switches and capacitor modules are used, and more costs are invested.

Therefore, for the realisation of HVNPG, if the voltage gain of the single-module circuit is greatly improved, it can not only reduce the demand for DC charging power supply but also reduce the number of modules and reduce the cost of switches.

Many papers have proposed the boost structure to improve the output pulse voltage gain [12–14]. Pulse transformers are often introduced together with primary pulse generators [15], such as Marx generators, to boost the voltage and get higher voltage gain [16]. The generator in Ref. [17] is compact consisting of a six-stage Marx generator and an HV pulse transformer, which increases the pulse amplitude by increasing the number of modules and the transformer ratio. However, due to the leakage inductance and distributed capacitance in the transformer, it is difficult to output pulses with a pulse width of ten nanoseconds. In addition, with the increase of voltage amplitude and frequency, magnetic saturation will obviously affect the quality of the pulse waveform.

It is mentioned in refs. [18–20] that the inductor is used as the secondary energy storage element to discharge pulses on the load through the cooperative action of the switch. The pulse amplitude obtained on the load will be higher than that on the primary energy storage unit so as to get a higher voltage gain.

In ref. [21], a solid-state Marx circuit using inductive energy storage is proposed. Inductance is added to each stage of Marx as the energy storage element and charged by the primary energy storage element capacitor. With switches turning off, inductances discharge in series to produce pulse on load. The four-stage generator can generate pulse with a voltage of 9.2 kV and pulse width of 170 ns, when the charging voltage is 200 V. However, the topology uses three switches series in a single stage, and the number of switches is not reduced compared to the traditional Marx circuit. In addition, because the pulse width is determined by the discharge time of the inductive load, the pulse width is difficult to achieve tens of nanoseconds.

An improved Blumlein line combined with an LC oscillator boost circuit was proposed in ref. [22]. The current in the inductor and the transmission line equivalent capacitance will produce LC-underdamped oscillation. The switch action at the high point of the oscillating voltage and pulse with the amplitude of twice the charging voltage, width of 5 ns, will be generated. However, limited by the switch withstand voltage and circuit topology, the maximum pulse amplitude is only 500 V. In addition, the voltage gain can be further improved.

In this paper, a new type of boost pulse-forming line generator is proposed. Combining the advantage of a short pulse generated by the transmission line and the principle of inductance boost, transmission lines are used as secondary inductive energy storage units to generate a high-voltage gain pulse. In addition, the application of variable impedance transmission line transformers (VITLT) multiplies the pulse voltage. An interesting fact is that the pulse generator can be modularised by the isolation effect caused by the transmission line time delay. A nanosecond pulse with extremely high-voltage gain is eventually generated, while the number of switches is reduced.

This paper is organised as follows. Section 2 describes the structure of the proposed new topology. The function of each subunit and the formation process of a high-voltage gain pulse are presented. In Section 3, the main consideration for the design of the generator is provided. To verify the theoretical analysis and topological principles, experimental research is carried out in Section 4. Finally, the characteristics of the generator are discussed and conclusions are drawn in Section 5 and Section 6, respectively.

2 | TOPOLOGY AND PRINCIPLE

2.1 | Topological composition

The topology of HVNPG is composed of three parts: pulse-forming part, voltage boost part and isolation part. It is shown in Figure 1.

In the pulse-forming part, capacitance is applied for the primary energy storage element which is parallel with DC charging power supply (U_{DC}). The transmission line ($Z_{storage}$) is applied for the secondary energy storage element. MOSFET is used for the pulse power switch (M_0). The variable impedance transmission line transformer (VITLT) is applied for the voltage boost part, which is cascaded after the pulse formation part. It consists of a single transmission line wound with a magnetic core. A longer transmission line (Z_{delay}) serves as the isolation part of the topology.

2.2 | Working principle

The following four stages provide a detailed description of the working principle of the proposed topology.

2.2.1 | Stage I (energy storage)

In this stage, a triggered on-state signal is applied to the gate of MOSFET, and the equivalent charging circuit is shown in Figure 2. R_z and L_z are the equivalent line resistance and inductance of $Z_{storage}$, respectively. R_d is the on-state resistance of MOSFET.

The equivalent inductance in the transmission line is charged by U_{DC}. The charging current is as follows:

$$i(t) = \frac{U_{DC}}{R_z + R_d}\left(1 - e^{-\frac{t}{\tau}}\right) \tag{1}$$

where $\tau = L_z/(R_z + R_d)$.

It can be seen from Formula (1) that in the case of selected MOSFET and transmission lines, the charging current is mainly determined by the charging voltage U_{DC} and the

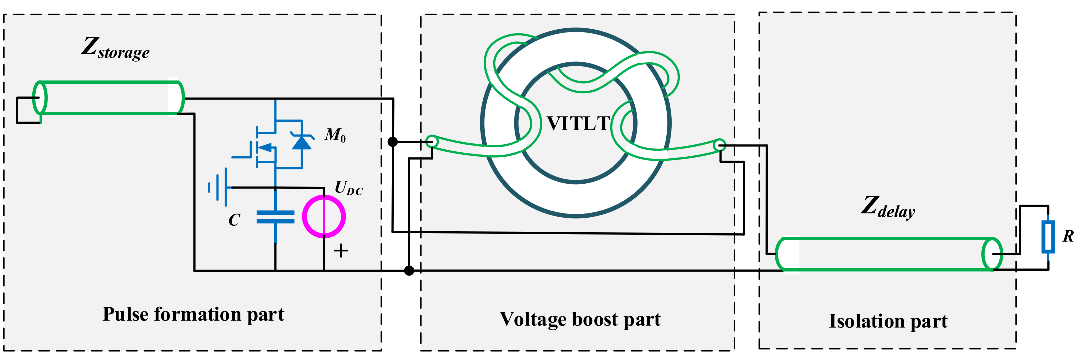

FIGURE 1 Schematic circuit diagram of a new type of the boost pulse-forming line generator.

charging time t. Thus, the charging current can be precisely controlled. And during the charging process, energy is stored in the transmission line in terms of the magnetic field.

2.2.2 | Stage II (pulse formation)

After charging time of Δt, the current in $Z_{storage}$ reached i_0. A triggered off-state signal is applied to the gate of MOSFET. And meanwhile, the energy stored in the $Z_{storage}$ is released. The equivalent circuit is shown in Figure 3. Z_{in} is the equivalent resistance of the pulse-forming circuit output port.

It can be regarded that the current flow of the M_0 branch is transferred to the Z_{in} branch. And when the switch is completely off, the current on the Z_{in} branch reaches its maximum value, that is, the pulse voltage reaches its peak. Therefore, the pulse rise time is determined by the turn-off time of the main switch.

Figure 4 shows the timing logical control and schematic diagram of the wave process. U_{gs} is the gate drive signal of M_{main}. l_T is the propagation time of $Z_{storage}$.

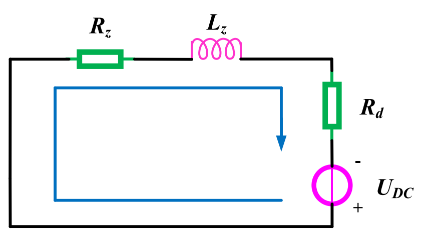

FIGURE 2 Equivalent charging circuit.

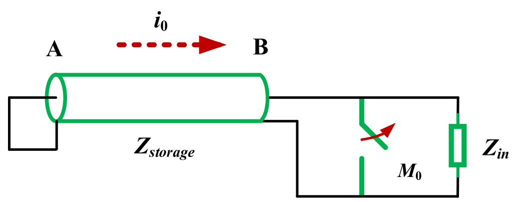

FIGURE 3 Equivalent pulse formation circuit.

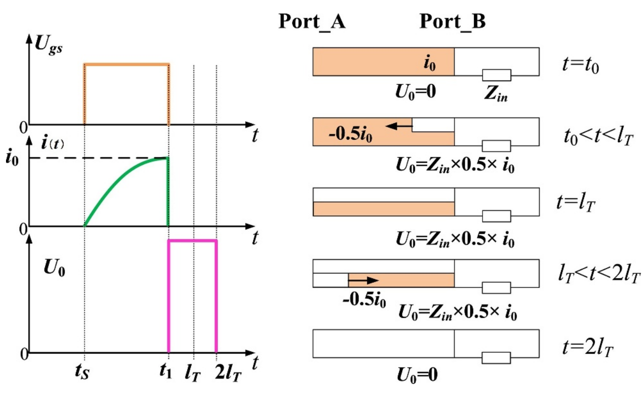

FIGURE 4 Timing logical control and schematic diagram of the wave process.

After M_0 turned off at t_1, the Port_B of the transmission line $Z_{storage}$ switches from the open state to the load matching state. The current wave i_b is generated at Port_B and travels to Port_A.

$$i_b = -\frac{Z_{in}}{Z_{storage} + Z_{in}} \cdot i_0 \tag{2}$$

At $t = l_T$, the current wave i_b arrives at Port_A. Since Port_A is in the short-circuit state, the current wave i_b is fully reflected and travels in reverse. Then, at $t = 2l_T$, the current wave i_b return to Port_B. Eventually, a pulse is generated at Port_B whose pulse width is $2l_T$ and amplitude is U_0. Since l_T is determined by the length of the transmission line, the pulse width is also determined by the length.

$$U_0 = Z_{in} \cdot \frac{Z_{storage}}{Z_{storage} + Z_{in}} \cdot i_0 \tag{3}$$

Under the condition of impedance matching, the ratio coefficient is 0.5. And the energy in $Z_{storage}$ is completely released during $2l_T$.

The output voltage mainly depends on the charging current of the pulse formation part. In specific applications, we usually choose a fixed charging time ($t > \tau$) and adjust the charging voltage to obtain the desired output pulse voltage. And the output voltage gain is as follows.

$$M_0 = \frac{U_0}{U_{DC}} = \frac{Z_{storage}}{2(R_z + R_d)} \cdot \left(1 - e^{-\frac{\Delta t}{\tau}}\right) \tag{4}$$

2.2.3 | Stage III (voltage boost)

VITLT is used for voltage boost after pulse generating from the forming unit. A mathematical model of VITLT is built. And the equivalent circuit is shown in Figure 5. The pulse-forming unit can be regarded as a pulse source with an internal resistance of $Z_{storage}$. U_1 and U_2 are the input and output port voltages of VITLT, respectively. Z_0, L_0 and l are the characteristic impedance, the equivalent inductance and the propagation time of the VITLT transmission line, respectively. I_1 and I_2 are the currents in the two equivalent branches of VITLT, respectively.

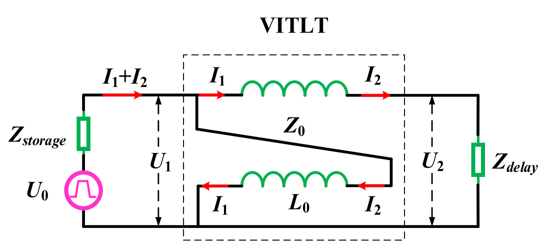

FIGURE 5 Equivalent circuit of the voltage boost part.

According to the theory of the transmission line, the following equation can be obtained.

$$U_1 = (U_2 - U_1) \cdot \cos \beta l + \mathrm{j} I_2 Z_0 \sin \beta l \tag{5}$$

$$I_1 = I_2 \cos \beta l + \mathrm{j} \frac{U_2 - U_1}{Z_0} \cdot \sin \beta l \tag{6}$$

where

$$\beta = \omega \sqrt{LC} = \frac{2\pi}{\lambda}$$

According to Kirchhoff's laws,

$$E = (I_1 + I_2) \cdot Z_{storage} + U_1 \tag{7}$$

$$U_2 = I_2 \cdot Z_{delay} \tag{8}$$

Simultaneous Equations (4) (5) (6) and (7), the output current and power expressions can be solved as follows.

$$I_2 = \frac{U_0 (1 + \cos \beta l)}{\varphi + \mathrm{j} \phi \sin \beta l} \tag{9}$$

$$P_2 = |I_2|^2 \cdot Z_{delay} = \frac{U_0{}^2 \cdot Z_{delay} (1 + \cos \beta l)^2}{\varphi^2 + \phi^2 \sin^2 \beta l} \tag{10}$$

where

$$\varphi = 2 Z_{storage} \cdot (1 + \cos \beta l) + Z_{delay} \cdot \cos \beta l$$

$$\phi = \frac{Z_{storage} \cdot Z_{delay}}{Z_0} + Z_0$$

From this expression (9), the conditions for maximum power transmission are obtained when βl is ideally much less than 1. And the maximum output power is as follows.

$$P_{2\_\max} = \frac{U_0{}^2}{4 \cdot Z_{storage}} \tag{11}$$

when

$$Z_{delay} = 4 \cdot Z_{storage} \tag{12}$$

In this condition, VITLT achieves 1:4 impedance transformation. According to the conservation of power, the output voltage U_2 is twice the input voltage U_1. And after voltage multiplying, the output voltage gain is as follows.

$$M_1 = 2M_0 = \frac{Z_{storage}}{R_z + R_d} \cdot \left(1 - \mathrm{e}^{-\frac{\Delta t}{\tau}}\right) \tag{13}$$

2.2.4 | Stage VI (time delay isolation)

To achieve higher voltage pulse output, a multi-stage superposition of a single-module circuit is required. Since the primary and secondary sides are directly interconnected, the proposed VITLT has no voltage isolation ability, which makes it impossible to stack multiple modules directly.

In this paper, a longer transmission line Z_{delay} is placed between the pulse-forming part and the load, and the isolation effect will be generated due to the delay effect. Figure 6 shows the schematic diagram of the n-stages transmission line time isolation method. n is the number of modules stacking. The pulse formation part of each module is on common ground and isolated from the load by a delayed transmission line. After travelling through the transmission line of equal length, synchronous voltage pulses are applied on the load at the same time to achieve voltage superposition.

For multi-module stacking, the non-synchronous of each module will become the influential factors, affecting the rise time and pulse width. In particular, the synchronisation of the switch action is crucial.

After multi-module superposition, the voltage gain is further improved.

$$M_2 = nM_1 = n \cdot \frac{Z_{storage}}{R_z + R_d} \cdot \left(1 - \mathrm{e}^{-\frac{\Delta t}{\tau}}\right) \tag{14}$$

3 | MAIN CONSIDERATION FOR DESIGN

3.1 | Design of impedance matching

In the new topology, transmission lines are used for energy storage, pulse forming, and pulse transmission. VITLT is used for pulse voltage multiplication, which involves impedance transformation. Therefore, the impedance of the transmission line used in different circuit units is also inconsistent. The impedance matching of each part is crucial in the design of the proposed generator. Otherwise, there will be a refraction and

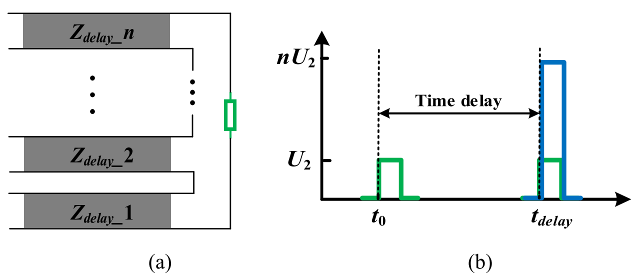

FIGURE 6 Schematic diagram of the n-stage transmission line time isolation method. (a) Circuit diagram of interconnection. (b) Diagram of pulse waveform.

reflection wave process inside the transmission line, affecting the pulse obtained on the load. For VITLT, the input impedance and output impedance are as follows.

$$Z_{in} = \frac{Z_{VITLT}}{2}\left[\frac{(\cos\beta l+1)^2 + \mathrm{j}\sin\beta l(1-\cos\beta l)}{(\cos\beta l+1)^2+\sin^2\beta l}\right] \tag{15}$$

$$Z_{out} = 2Z_{VITLT}\cdot\left[\frac{(\cos\beta l+1)^2 - \mathrm{j}\sin\beta l(1-\cos\beta l)}{(\cos\beta l+1)^2+\sin^2\beta l}\right] \tag{16}$$

when the length of transmission line is much smaller than the wave length, that is, $l << \lambda$, $\sin\beta l \approx 0$. Impedance characteristics of VITLT transmission line can be calculated using the following equation.

$$Z_{VITLT} = \sqrt{(Z_{in}\cdot Z_{out})} \tag{17}$$

Figure 7 shows the schematic diagram of the impedance matching design in the proposed generator.

It can be seen that the three kinds of characteristic impedances of the transmission line, including $Z_0/2$, Z_0, and $2Z_0$, are required.

Because the characteristic impedance of commercial coaxial transmission lines is mostly 50 Ω or 100 Ω, we use a combination of 25, 50 and 100 Ω transmission lines to implement the proposed pulse generator. The transmission line with a characteristic impedance of 25 Ω can be constructed in parallel by two 50 Ω transmission lines.

Since the impedance conversion ratio from the VITLT's primary to secondary side is 1:4, 25 Ω is used as the energy storage transmission line impedance $Z_{storage}$ and 100 Ω as the impedance of isolated transmission line Z_{delay}. As can be seen from formula (17), the characteristic impedance of the transmission line constituting VITLT is 50 Ω. The output impedance of the entire single-module generator can be seen as 100 Ω. Thus, the matched load impedance is 100 Ω.

3.2 | Design of VITLT

VITLT is made by winding a transmission line around a magnetic core, whose energy transmission is realised using the transmission line distribution parameters. This is different from the energy transfer method of traditional transformers through magnetic flux coupling. According to the analysis in Section 2, the optimal power transmission condition is that the transmission line length l is much smaller than the wave length λ on the transmission line. Formula (18) is the general selection principle in engineering.

$$l < \frac{1}{8}\lambda = \frac{c}{f\sqrt{\varepsilon_r}} \tag{18}$$

It can be seen from Figure 5 that the pulse generated from the pulse-forming part enters into the transmission line and is also added to the equivalent inductance L_0 of VITLT. If L_0 is too small, a portion of the pulse energy will flow back through the inductance to the pulse-forming unit. Eventually, the pulse energy obtained on the load is reduced. Therefore, the transmission line length l should not be too short. In addition, the winding magnetic core can greatly increase the inductance, thus blocking the energy backflow. L_0 can be determined by the following formula.

$$L_0 = \frac{\mu_r\cdot\mu_0\cdot N^2\cdot A_w}{l_w} \tag{19}$$

where A_w is the magnetic circuit cross-sectional area of the core, l_w is the length of a magnetic path, μ_0 and μ_r are the vacuum permeability and the core's relative permeability and N is the turns' number of the transmission line around VITLT.

The leakage current of the inductive loop is as follows.

$$i_w = \frac{U_1}{L_0}\cdot\left(1-\mathrm{e}^{-\frac{T_w}{\tau_0}}\right) \tag{20}$$

where

$$\tau_0 = \frac{L_0}{R_{Z_0}}$$

In engineering design, the leakage current is always less than 1 mA. And the core parameters and turns can be calculated from this. In addition, for nanosecond pulses with a specific amplitude and pulse width, the core parameters should also meet the following formula.

$$\int_0^{T_w} U_1 dt \le NA_w B_s \tag{21}$$

where T_w is the pulse width and B_s is the core saturation flux.

4 | EXPERIMENTAL VERIFICATION

To verify the principle of topology, a single-module experiment was conducted, and the prototype of a five-stage stacker was built in this paper. Table 1 shows the experimental parameters.

A signal generator modelled AFG31000 was used to generate TTL signals, and after optical isolation, MOSFET gate drive signals were provided. The model of the oscilloscope

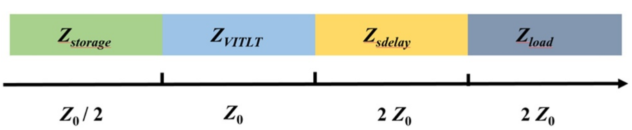

FIGURE 7 Schematic diagram of the impedance matching design.

is WavePro 760Zi-A (LeCroy, 6 GHz). The DC charging power supply is KA6003D (KORAD, 60 V). The load uses thick film non-inductive resistors with a very low stray inductance. All transmission lines used are double-layer shielded lines, with the aim of reducing spatial interference between lines. Nanocrystalline materials with high initial permeability are used as the core of VITLT.

As for the measurement of pulse voltage, a combined measurement system, consisting of a passive voltage probe and resistance divider, was utilised to measure voltage pulses over 10 kV. The model of the passive voltage probe is PPE6 kV (LeCroy, 400 MHz) with an attenuation ratio of 1:100. The resistance divider is self-made integrated on the PCB board whose attenuation ratio is 1:2. They are interconnected through specific connectors. The combined attenuation ratio is 1:200, and the voltage pulse of 12 kV can be measured. Figure 8 shows the diagram of the pulse voltage measurement system. The current probe modelled CP031 (LeCroy, 100 MHz) was used for current measurement.

4.1 | Single-module experiment

In this paper, a single-module prototype is built first to verify the circuit principle and boost characteristics. This paper builds a single-module prototype, as shown in Figure 9. In the experiment, the signal generator trigger pulse width is set to 2 μs. This means that the charge time of the energy storage pulse formation line $Z_{storage}$ is 2 μs. During the charging time, set the voltage of the DC supply to 20 V.

TABLE 1 Experiment parameters and values.

Name	Parameter
M_{main}	G3R75MT12 J
C	100 μF
Length of $Z_{storage}$	1 m
Length of Z_{delay}	3 m
Length of Z_{VITLT}	12 cm
$Z_{storage}/Z_{VITLT}$	SFX-50
Z_{delay}	CSLFF-100
Magnetic core	Nanocrystalline
Load impedance for a single-module circuit	100 Ω
Load impedance for the five-stage stacker	500 Ω

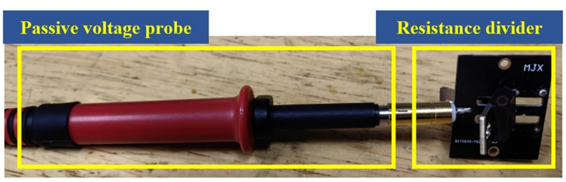

FIGURE 8 Diagram of the pulse voltage measurement system.

Figure 10 shows the current on the single energy storage transmission line after the charging time of 2 μs. It can be seen that the circuit is in the energy storage stage from the start time of triggering MOSFET conduction. And the current in the energy storage transmission line keeps increasing, whose trend of change is consistent with the description in Equation (1). After 2 microseconds of charging, the current reaches 41.5 A. Since two identical transmission lines are connected in parallel as $Z_{storage}$, the charging current doubles to 83 A at this time. After the MOSFET is triggered to turn off, the current drops rapidly to about 0 A.

Figure 11 shows the voltage on the VITLT's primary and secondary side. Pulse on the primary side with amplitude of 1 kV, pulse width and rise time of 10 and 3.2 ns are generated using the pulse forming part. Voltage on the secondary side of the VITLT becomes 2 kV. The pulse width and rise time are 10.8 and 3.5 ns, respectively. From the amplitude of the pulse voltage, VITLT can achieve a double voltage boost. Comparing both the pulse rise time and width, there is not much difference, which means that the loss of a high-frequency component of the pulse is very small after the VITLT boost.

The single-module experiment verifies the topology principle and the boost effect of VITLT on a short nanosecond pulse. The voltage gain of the single module can reach 100. And the MOSEFET used withstands 1.2 kV and achieves a pulse voltage output of 2 kV, theoretically saving one switch.

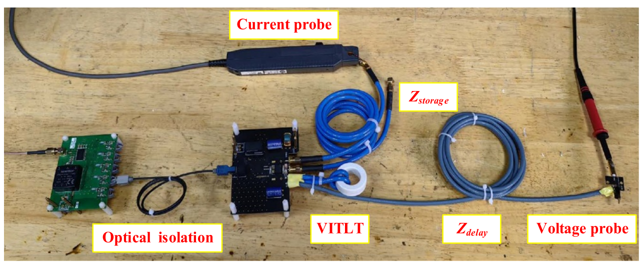

FIGURE 9 Diagram of each part of the single-module circuit.

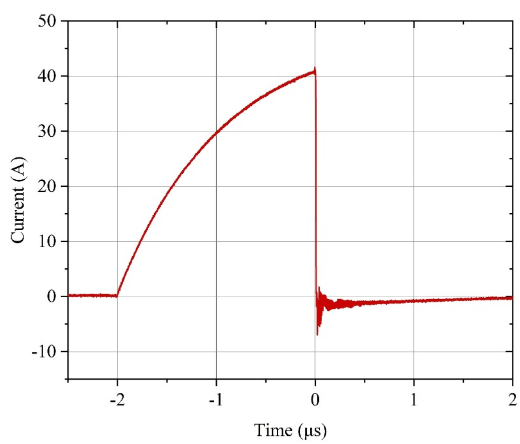

FIGURE 10 Current on the single energy storage transmission line.

4.2 | Multi-modules stacking experiment

Due to the advantage of easy operation and maintenance, strong versatility, conducive to standardised production, modular pulse stacking is one of the main trends in the development of all solid-state pulse power source technology in recent years. The topology proposed in this paper can also be modularly stacked to achieve a high-voltage pulse. This paper builds a 5-stage module prototype to verify the superposition principle. Figure 12 shows the experimental platform.

The five-stage stacker has a matching load impedance of 500 Ω. To measure higher voltages with a 6 kV probe, we use two 250 Ω resistors in series as the load and measure it through voltage division. The switching power supply is used to supply power to the main circuit PCB.

Figure 13 shows the waveform of the pulse voltage on the load. A 10-kV voltage pulse is generated using the 5-stage stacker with a pulse width and rise time of 12 and 4.2 ns, respectively. Compared with the single-module experiment, the waveform effect after superposition has some changes. It can be seen that the pulse width and rise time are increased compared with the results of the single-module experiment. The reason is the dispersion of each module and the asynchronicity of the MOSFET gate drive signal. At this time, the charging voltage of the pulse generator is only 28 V. And the voltage gain can reach 357.

Compared with ref. [23], due to the lack of further theoretical analysis and optimised design of the boost part in ref. [23], the output waveform distortion of the single module is serious, and the flat top has disappeared. In addition, the 7-stage module with each output of 1 kV generates voltage pulses of 10.7 kV, and the superposition efficiency is extremely low.

Figure 14 shows the current on the single energy storage transmission line in the 5-stage stacker. It can see that the current increases from 41.5 to 47.2 A compared to the single-module test. This indicates that the superposition efficiency is not 100% when applying the time delay isolation method to modular superposition. The energy stored in the transmission line can be obtained by the following formula when the charging current reaches i_0.

$$W_{storage\_n} = 2n \cdot \frac{1}{2} L_Z \cdot {i_{0\_n}}^2 \tag{22}$$

where n is the number of modules superimposed.

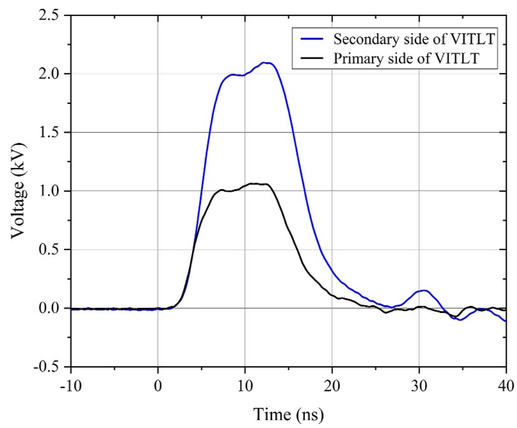

FIGURE 11 Voltage on the primary and secondary side of VITLT.

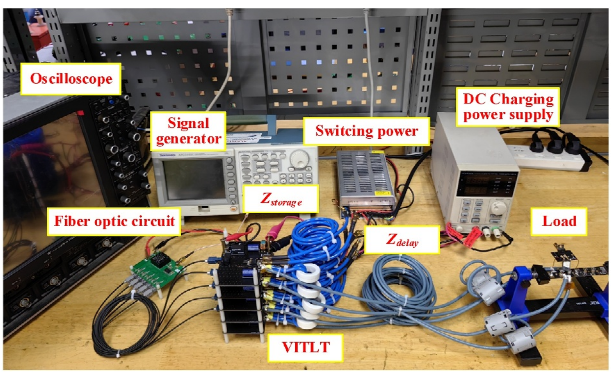

FIGURE 12 Test platform of 5-stage stacking.

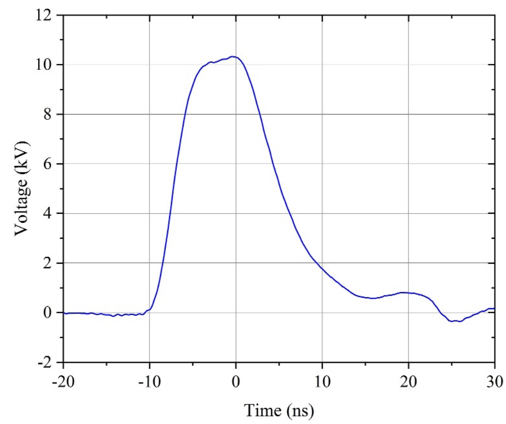

FIGURE 13 Output waveform of the 5-stage stacker.

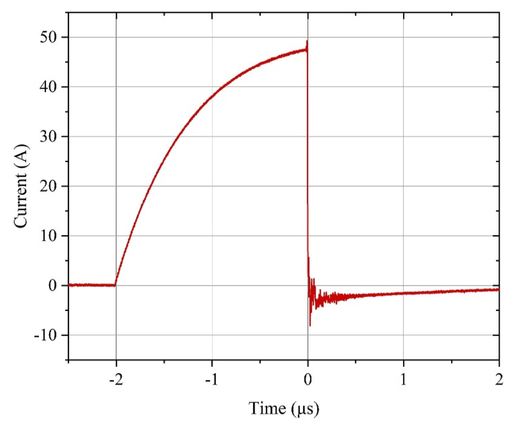

FIGURE 14 Current on the single energy storage transmission line.

$$\eta = \frac{W_{storage\_1}}{W_{storage\_5}} = \left(\frac{i_{0\_1}}{i_{0\_5}}\right)^2 \tag{23}$$

It can be calculated from formula (24) that the superposition efficiency of 5-stage modules stacking applying the time delay isolation method is 77.3%.

Finally, we tested the stability of the refrequency operation of the 5-stage stacker. Figure 15 shows the result of running at 100 kHz in burst mode.

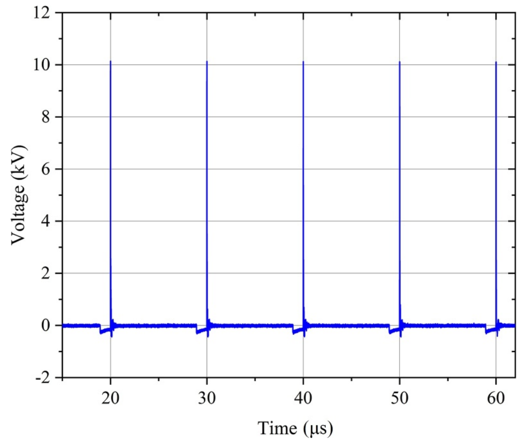

FIGURE 15 Output pulse waveform of 100 kHz operation.

5 | DISCUSSION

The proposed generator combines the inductive energy storage of transmission lines with a variable-impedance transmission line transformer to generate a nanosecond pulse with high-voltage gain. The isolation effect caused by transmission line delay has been applied to achieve modular stacking. We had built a prototype for experimental research and verified the circuit principle and boost effect.

Many papers have also proposed methods to improve pulse voltage gain. The authors in ref. [17] use a pulse transformer to boost voltage, but it is difficult to achieve a short pulse of tens nanoseconds. The authors in ref. [21] or [22] introduce inductors as intermediate energy storage elements to generate high-gain pulses. But limited by circuit methods, the gain has not increased too much and still requires high-voltage charging of several hundred kilovolts.

And the comparison with the prototype built in this paper is shown in Table 2. It can be seen that the prototype in this paper has extremely high-voltage gain and uses fewer switches. Charging voltage with several tens of volts can be achieved only by a low-voltage digital power supply.

However, the time isolation method applied in the proposed topology has superposition loss, that is, non-linear power accumulation. When the accumulated voltage amplitude is too high, there is a backflow of energy between modules, which causes oscillations at the tail of the pulse waveform. A more efficient stacking method for research is also an upcoming work to be carried out in the future.

TABLE 2 Comparison between different prototypes.

Prototypes	[17]	[21]	[22]	This paper
Pulse parameter	35 kV/ 250 ns	9.2 kV/ 120 ns	500 V/ 5 ns	10 kV/ 12 ns
Number of switch	6	12	2	5
Number of stage	6	4	1	5
Charging voltage	850 V	200 V	250 V	28 V
Voltage gain	41.2	46	2	357

6 | CONCLUSION

The paper proposed a new type boost nanosecond pulse generator with extremely high-voltage gain. The principle of inductance energy storage for the transmission line and the method of a variable-impedance transmission line transformer boost are proposed to achieve higher voltage gain. Numerical modelling analysis, parameter design and experimental verification are carried out, respectively, for the proposed topology. The following conclusions can be drawn:

1) When the distributed inductor of the transmission line is used as the energy storage unit, nanosecond pulses with high-voltage gain can be generated, whose pulse width is determined by the length of the transmission line.
2) The proposed variable-impedance transmission line transformer can multiply the voltage amplitude of a nanosecond pulse and ensure the integrity of the pulse waveform after the voltage boost.
3) The isolation effect caused by transmission line delay can make the proposed topology realise modular stacking. The prototype test shows that when only charging voltage of 28 V is provided, the five-module stacker prototype can output a pulse with a voltage amplitude of 10 kV and pulse width of 12 ns. And the voltage gain can reach 357. The demand for DC charging power can be greatly reduced.

ACKNOWLEDGEMENTS

This work was supported by the National Natural Science Foundation of China (52277135 and 52237010) and the Graduate Student Research and Innovation Project of Chongqing (No. CYB23027).

CONFLICT OF INTEREST STATEMENT

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Ma, J., et al.: 10 kV nanosecond pulse generator with high-voltage gain and reduced switches. High Voltage. 9(5), 1059–1067 (2024). https://doi.org/10.1049/hve2.12470